

Qatar-1b

R-0637

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230825 · created 2026-07-25T18:02:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460181.76648 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.012
Transit depth (Rp/Rs)^2	3.67 +/- 0.47 [%]
Orbital Inclination (inc)	81.75 +/- 0.31
Airmass coefficient 1 (a1)	1.019 +/- 0.011
Airmass coefficient 2 (a2)	-0.015 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.58
Optimal Annulus	14.1
Transit Duration (day)	0.0574 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.012 vs 0.1463 (deviation 3.15 σ)
Duration fit vs published	0.0574 d vs 0.0692 d (deviation 3.04 σ)
Mid-time O-C	-6.6 min
Depth SNR	13.2
Residual scatter	1.09 %

Light curve

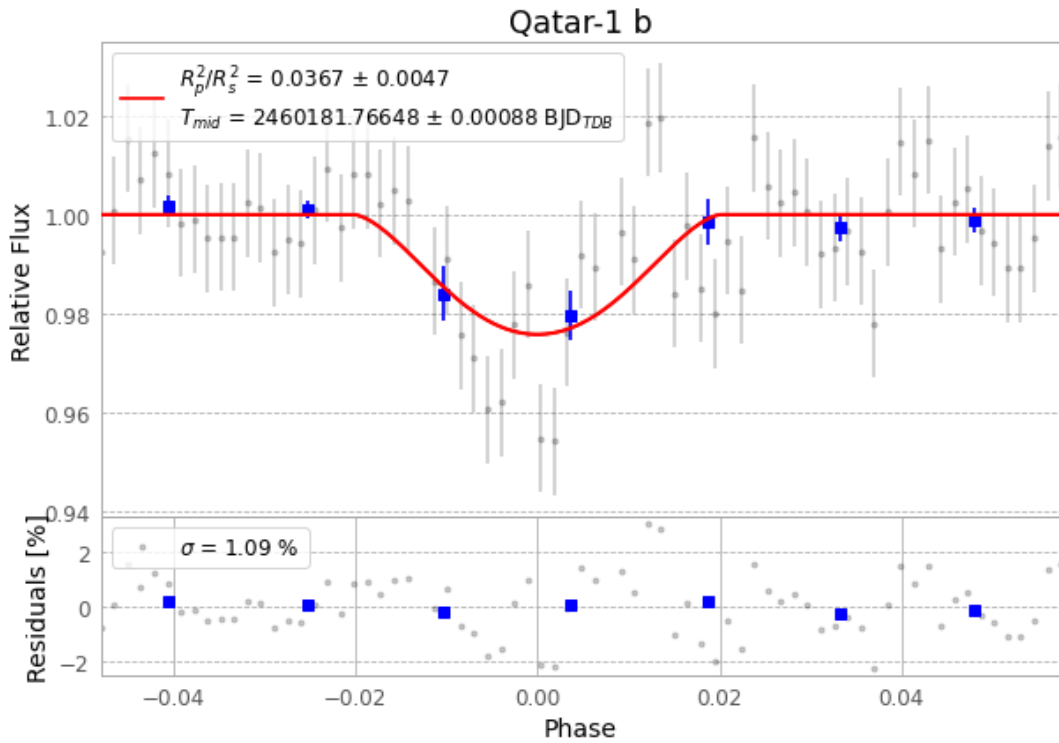


Figure 1 — FinalLightCurve_Qatar-1 b_2023-08-24.png (SHA-256 c7023adff7ad05b4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 191], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 720eb9a9933bd151...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

73 FITS frames (48 MB), Qatar-1230825044218.FITS → Qatar-1230825081816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230825.log (9662a666eddc...), FinalLightCurve_Qatar-1 b_2023-08-24.png (c7023adff7ad...), FinalLightCurve_Qatar-1 b_2023-08-24.csv (d9f564bc82c7...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 18:02Z.